

1. Real Dialogue Input

A real chat session from U-Sticker dataset

Current session (target turn)

Alice this is good

Bob I would prefer this be more common because I run into this exact thing quite often

Alice yeah

Bob i just use a flag because it's more obvious to me

Alice well... 

History (previous sessions)

- Session A: shared a quote about TF2 engineers
- Session B: mentioned confusing F-Droid with F-List
- Session C: detections of a link and offensive content
- ...

→ used for memory retrieval

2. Information Available for the Task

Information we can use to select the best sticker



Local Context (**context_text**)

Last turns before the target sticker

Bob: I would prefer this be more common because I run into this exact thing quite often
Alice: yeah
Bob: i just use a flag because it's more obvious to me
Alice: well...



Retrieved Memory (**memory_text**)

Top-3 relevant historical session memories

- #1 A user shared a quote about TF2 engineers.
- #2 Another mentioned confusing F-Droid with F-List.
- #3 There were detections of a link and offensive content.



Reply Intent (**intent_text**)

LLM-generated description of the reply intent

humor_teasing: The reply is a playful joke, likely a sticker with a humorous face or a funny meme.



Sticker Retrieval Model

rank stickers from candidate bank

3. Ranked Sticker Outputs

Model output: a ranked list from the sticker bank

Observed reply (ground truth)



Top-5 Ranked Stickers (from sticker bank)



1



...

2



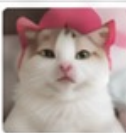
...

3



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4



...

5



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